CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
An Autonomous Institute I Affiliated to Osmania University
Kokapet Village, Gandipet Mandal, Hyderabad, Telangana-500075, www.cbit.ac.in

COMMITTED TO RESEARCH, INNOVATION AND EDUCATION 48 years

COURSE-END PROJECT REPORT

On

SOCIAL MEDIA ENGAGEMENT ANALYSIS

Submitted to fulfill the requirements for the Award of CIE Marks (10)

B.E. (IT), III-SEM | Exploratory Data Analysis & Visualization (EDAV - 22ADC31N)

By

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A.Y. 2026–2027



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CERTIFICATE

This is to certify that the course-end project report entitled “Social Media Engagement Analysis” is a record of the work carried out by Y.Hansika (1601-25-737-134), in partial fulfilment of the requirements of the course Exploratory Data Analysis and Visualization (22ADC31N) of the B.E. (IT), III-SEM programme during the academic year 2026–2027.

The project examines a social-media engagement dataset through Python-based data inspection, cleaning, statistical summaries, aggregation, correlation analysis, time-series analysis and visualization. The report is prepared with reference to the submitted notebook and its recorded outputs.

Place: Hyderabad

Date: ____ / ____ / _____

Course Faculty
(Mr. Geedikanti Shrikanth)



DECLARATION

I hereby declare that the project report entitled "Social Media Engagement Analysis" submitted to the Department of Information Technology, CBIT, is a record of work carried out by me for the course Exploratory Data Analysis and Visualization under the guidance of the course faculty, Department of IT, CBIT.

The report documents the dataset fields, data-quality checks, descriptive analysis, group-wise comparisons, correlation analysis, time-series analysis and visualizations. The interpretations are limited to the results supported by the submitted notebook and are not intended to establish causal relationships.

Place: Hyderabad

Date: ____ / ____ / _____

Signature of the Student
Y.Hansika | 1601-25-737-134

ACKNOWLEDGEMENT

I express my sincere gratitude to the faculty Mr. Geedikanti Shrikanth, of the Department of Information Technology, Chaitanya Bharathi Institute of Technology, for providing guidance and an academic environment in which to complete this course-end project.

I acknowledge the developers and maintainers of Python, NumPy, Pandas, Matplotlib and Seaborn. These open-source tools support the inspection, processing, summarization and visualization of data.

I also acknowledge the supplied Social Media Engagement dataset and the submitted notebook, which form the basis of this report. The notebook outputs are used as the primary evidence for the numerical findings described in the project.

Y.Hansika | 1601-25-737-134

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ABSTRACT

Social media platforms generate large volumes of activity data, including likes, shares, comments, impressions, sentiment, toxicity indicators, campaign phases and engagement rates. This project applies exploratory data analysis (EDA) to a dataset containing 12,000 social-media posts and 28 columns. The analysis is implemented in Python using NumPy, Pandas, Matplotlib and Seaborn in a Google Colab/Jupyter notebook.

The project compares five platforms—Facebook, YouTube, Reddit, Instagram and Twitter—and investigates brand-level performance, sentiment groups, campaign phases and changes over time. Data preparation includes checking missing values and duplicates, converting timestamps to datetime format, imputing missing numeric values with medians and missing categorical values with "Unknown," and deriving an engagement category. Grouping, aggregation, correlation matrices, daily and monthly summaries, and multiple plots are used to explore patterns.

The recorded outputs show Facebook has the largest total likes (6,133,107) and the highest reported average engagement rate among platforms (approximately 0.29, rounded). Microsoft has the highest average engagement rate among the listed brands (0.341), while Samsung has the largest total likes (3,073,984). The date with the highest daily average engagement rate in the notebook is 19 May 2024, with an average of 1.6561 across 18 posts. The correlation matrix shows a negative association between impressions and engagement rate (−0.30), while most other displayed linear correlations are close to zero.

1.INTRODUCTION

Social media is a channel for communication, entertainment, customer support, public discussion and brand promotion. Social media is a channel for communication, entertainment, customer support, public discussion and brand promotion. Organisations use networks to publish content reach audiences and observe how users respond. Organisations use networks to publish content reach audiences and observe how users respond. However the number of posts and the volume of interaction make it difficult to understand performance by looking at records. However the number of posts and the volume of interaction make it difficult to understand performance by looking at records. Exploratory data analysis provides a way to summarize such activity and identify patterns that deserve further investigation. Exploratory data analysis provides a way to summarize such activity and identify patterns that deserve further investigation.

Engagement can be represented through interactions such as likes, shares and comments. Engagement can be represented through interactions such as likes, shares and comments. Impressions provide a measure of how content was displayed while an engagement-rate field attempts to express interaction relative to a denominator defined by the dataset. Impressions provide a measure of how content was displayed while an engagement-rate field attempts to express interaction relative to a denominator defined by the dataset. These measures describe aspects of performance: a post may receive many likes because it was widely viewed while another may receive fewer interactions but a stronger interaction rate. These measures describe aspects of performance: a post may receive many likes because it was widely viewed while another may receive fewer interactions but a stronger interaction rate.

This project uses the submitted notebook to explore engagement across Facebook, YouTube, Reddit, Instagram and Twitter. This project uses the submitted notebook to explore engagement across Facebook, YouTube, Reddit, Instagram and Twitter. It also examines brand names, campaign phases, sentiment labels, sentiment scores, toxicity scores and timestamps. It also examines brand names, campaign phases, sentiment labels, sentiment scores, toxicity scores and timestamps. The analysis is descriptive than predictive: it seeks to summarize what appears in the data not to claim that a platform, sentiment or campaign phase causes engagement. The analysis is descriptive than predictive: it seeks to summarize what appears in the data not to claim that a platform, sentiment or campaign phase causes engagement.

EDA is especially useful early in a data project because it can reveal distributions missing data, outliers, inconsistent types and relationships that may guide future hypotheses. EDA is especially useful early in a data project because it can reveal distributions missing data, outliers, inconsistent types and relationships that may guide future hypotheses. The charts and grouped tables in this report turn row-level data into comparisons that can be discussed and checked. The charts and grouped tables, in this report turn row-level data into comparisons that can be discussed and checked.

PROBLEM STATEMENT, OBJECTIVES AND SCOPE

Problem statement: We found that the dataset holds thousands of social-media post records and many engagement, content and context variables. Without processing it is hard to compare platform performance look at brand activity see engagement across sentiment groups check time-based changes or study relationships among numeric fields. The project tackles this issue with an EDA workflow.

Primary objectives:

• Inspect the dataset dimensions, columns, data types descriptive statistics missing values and duplicate rows.

• Summarize likes, shares, comments, impressions and engagement rate across platforms.

• Compare sentiment. Brand-level engagement measures.

• Examine performance by campaign phase. Investigate daily and monthly trends.

• Calculate correlations between selected variables.

• Communicate findings with bar plots, heatmaps, a pie chart, line plots, histograms, scatter plots and box plots.

Scope: The analysis covers the fields in the notebook. It does not independently verify user identities, platform APIs, paid campaign costs conversion outcomes or the business meaning of each engagement-rate value. It does not train a model perform causal inference or make claims, about all users of each platform. The dataset is treated as the working sample supplied for the course project.

REQUIREMENTS ANALYSIS AND TECHNOLOGY STACK

Here is the input from the user:

Hardware and software requirements include a computer with a browser, internet access for Google Colab if used, Python 3.x and the input CSV file named Social_Media_Engagement_Dataset.csv. The notebook uses data-science libraries that are widely available in Colab.

Python is the core programming language. Its readable syntax makes it suitable for data loading, transformation, aggregation and plotting. NumPy provides numerical arrays and functions such as mean minimum, maximum and random sampling. Pandas provides DataFrames, missing-value checks, duplicate removal, type conversion, groupby aggregation, pivot tables, merges, reshaping and time-series resampling.

Matplotlib supplies plotting primitives and figure formatting. Seaborn builds plots such as bar plots, heatmaps, histograms, scatter plots and box plots on top of Matplotlib. The notebook imports all four libraries. Performs the analysis interactively.

Functional requirements: load the CSV into a DataFrame; inspect its schema; check missing and duplicate records; prepare timestamps and missing values; calculate platform, sentiment, brand and campaign summaries; calculate a correlation matrix; derive /monthly engagement summaries; and render plots, with meaningful labels and titles.

Non-functional considerations include reproducibility, legible visualizations, transparent transformations and accurate interpretation. The notebook uses a fixed seed (42) for its 100-record sample, which allows that sample to be reproduced under the same data and library conditions.

DATASET SOURCE, STRUCTURE AND DATA DICTIONARY

The notebook reads a CSV file called Social_Media_Engagement_Dataset.csv. The recorded inspection output shows 12,000 rows and 28 columns before cleaning. The fields represent post identifiers, timestamps, platforms, content metadata, sentiment indicators, interaction counts, impressions, engagement rate, brands, products and campaign context.

Key identity and context fields: post_id identifies a post; timestamp records its time; day_of_week provides a weekday label; platform identifies the network; user_id is a pseudonymous user identifier; location and language describe user/content context.

Content fields: text_content stores post text; hashtags and mentions capture references; keywords and topic_category summarize subject matter. Brand_name, product_name, campaign_name and campaign_phase describe the campaign context.

Engagement fields: shares_count and comments_count count three interaction types; impressions records exposure; engagement_rate is a precomputed rate supplied in the dataset. The notebook does not define the formula used to create engagement_rate. Its precise denominator and scale should be verified before comparing it with external benchmarks.

Sentiment and safety-related fields: sentiment_score is numeric sentiment_label contains Positive, Negative categories, emotion_type records an emotion category and toxicity_score is a numeric indicator. Additional numeric fields include user_past_sentiment_avg, user_engagement_growth and buzz_change_rate.

Important scope note: this report describes the schema and recorded outputs in the submitted notebook. The CSV itself was not separately attached in this conversation so raw-record revalidation is, outside this reports evidence.

DATA FLOW AND PROJECT ARCHITECTURE

The workflow follows a sequence: input CSV → load into Pandas → inspect schema and quality → clean and transform → compute summaries → analyze relationships and time trends → visualize → interpret and document.

Stage 1 environment setup: import NumPy, Pandas, Matplotlib and Seaborn. The notebook calls the Google Colab file-upload workflow. Reads the named CSV into df.

Stage 2 inspection: display column names, DataFrame information and descriptive statistics. Check counts per column and count exact duplicate rows. This provides a baseline before transformations.

Stage 3 cleaning: drop rows; parse timestamp values using pandas.to_datetime with errors='coerce'; fill missing numerical fields with the median of each numeric column; and fill missing object/category fields with 'Unknown'. Re-check missing values and duplicates.

Stage 4 feature preparation: categorize engagement_rate into Low, Medium and High using cut points of 0.05 and 0.15; use an index based on platform and sentiment; reshape a small sample from wide to long format; and merge brand engagement summaries with brand post counts and average sentiment.

Stage 5 analysis: calculate NumPy statistics for likes, by platform; aggregate platform, sentiment, brand and campaign-phase metrics; calculate correlations; create monthly time series; and identify the date with the highest daily average engagement rate.

Stage 6 communication: create a variety of charts. Explain the observations. This staged workflow improves traceability because each result can be linked to a transformation or aggregation.

DATA CLEANING AND PREPARATION

Data cleaning is important because missing fields, duplicate records and wrong data types can mess up summaries. The notebook first looks at missing values and duplicate rows. The first result shows 3,941 missing values in mentions while other columns have no missing values. It also shows there are zero duplicate rows.

The cleaning step uses drop_duplicates() changes the timestamp to a format using errors='coerce' fills missing numbers with the median of each column and fills missing text columns with 'Unknown'. After cleaning the data still has 12,000 rows and 28 columns, no missing values and no duplicate rows.

Using the median to fill numbers works better with extreme values because the median is not as affected by very high or very low numbers as the mean. It is still a choice made by the person doing the analysis. Missing data might have meaning and replacing all numbers with a median can make the data less varied. For this dataset the missing data is in mentions so filling numbers might not change anything if no numbers are missing.

Changing the timestamp to a format allows for filtering by date grouping by day and resampling by month. Using errors='coerce' means that bad timestamp strings become values instead of stopping the code. Time-series work should check how many timestamps turned into missing values.

Outliers should not just be removed automatically. The highest engagement_rate is 32.21171 while the median is 0.080605, which shows a skewed distribution. There might be values or a problem, with the scale or how the rate is defined. These numbers should be checked against the datasets formula and original records before using the rate for decisions.

EXPLORATORY DATA ANALYSIS: OVERVIEW

EDA starts with dimensions, variable types, descriptive statistics and frequency counts. The notebook contains 12,000 records. The main numeric variables are sentiment_score, toxicity_score, likes_count, shares_count, comments_count, impressions, engagement_rate, user_past_sentiment_avg, user_engagement_growth and buzz_change_rate.

The average values recorded are: likes_count average is 2,490.72 shares_count average is 1,007.17 comments_count average is 504.35 impressions average is 49,811.34 and engagement_rate average is 0.278137. The range for likes goes from 0 to 5,000. Shares range from 0 to 2,000. Comments range from 0 to 1,000. Impressions range from 130 to 99,997.

The sentiment_score ranges from about −1 to +1 with a mean close to zero at 0.000553. The toxicity_score has a mean of 0.503868. Ranges from 0 to nearly 1. The user_engagement_growth has a mean near zero. The buzz_change_rate spans from −99.9 to +99.9.

These numbers help describe the center spread and range of the data.. They don’t explain why users interact with a post. A close to zero doesn’t necessarily mean no emotions. It could mean numbers of positive and negative reactions. Likewise a high average impression count doesn’t automatically mean engagement. A post can get views but still fail to get much interaction.

The summary, in the notebook is the first step. The next steps will split the data by platform, sentiment, brand, campaign phase and time. These breakdowns help reveal patterns that a single average number can hide.

QUESTION-WISE METHODOLOGY AND IMPLEMENTATION

I use the notebook to set up the analysis around five key questions. The main analysis is organized so that each question guides the step.

Q1: What are the average, minimum and maximum likes for each platform? I first gather the platform labels then slice the DataFrame for each platform. I convert likes_count into a NumPy array. Run np.mean() np.min() and np.max(). This shows how array conversion, filtering and descriptive statistics work together.

Q2: Which platforms and brands get the engagement and how does engagement shift with sentiment? I use Pandas groupby and agg to sum likes, shares, comments, average engagement rate and impressions for each platform. I then add groupings to show average interactions and engagement rate broken down by sentiment label and brand.

Q3: Are there missing values or duplicate records. How does the notebook handle them? I drop rows parse timestamps, fill missing numeric values with medians replace empty categories with 'Unknown' and then verify the cleaned dataset.

Q4: How does engagement shift across platforms and dates and what ties exist among the numbers? I group data by platform and campaign_phase calculate daily average engagement and post counts. I resample the data to averages spot the highest daily average and build a correlation matrix.

Q5: What stories do the bar plots, scatter plots, pie charts, line plots and heatmaps tell? I plot total likes and comments for each platform show engagement, by platform and sentiment map sentiment distribution display platform post share highlight brand likes visualise engagement-rate spread compare impressions to likes spread data and mark toxicity across sentiments.

The methodology stays descriptive and reproducible. I read the recorded outputs. Draw conclusions rather than guessing based on generic social media expectations.

Q1 — NUMPY STATISTICS OF LIKES BY PLATFORM

NumPy statistics are calculated by selecting the values for each platform and turning that selection into a NumPy array. Using np.mean gives the number of likes per record. Np.min and np.max show the highest values observed in the data.

Recorded outputs:

• Instagram: mean 2,506.01; minimum 1; maximum 5,000.

• Twitter: mean 2,445.19; minimum 1; 4,999.

• Reddit: mean 2,497.91; minimum 1; 5,000.

• YouTube: mean 2,481.82; minimum 4; 5,000.

• Facebook: mean 2,522.87; minimum 0; 5,000.

Facebook has the mean likes among the five platforms. After Facebook the order is Instagram and then Reddit. Twitter has the mean likes. The differences between the means are not very large when compared to the range of values seen on each platform.

The minimum and maximum values should not be taken as indicators of performance. A minimum of zero on Facebook means there is least one post with no likes. A maximum close to 5,000 shows the number of likes recorded in the dataset. Means can be affected by values so looking at medians and the full distribution of the data can provide more useful information.

This example shows how NumPy can work with Pandas. Pandas is used to filter the rows based on platform. Then NumPy takes the filtered data. Performs numerical calculations. It is a starting point, for comparing engagement across different platforms later.

Q2 — PLATFORM-WISE ENGAGEMENT ANALYSIS

The platform summary brings together likes, shares, comments, average engagement rate and total impressions. The table that is recorded is arranged based on likes.

Platform totals: Facebook has 6,133,107 likes, 2,388,476 shares, 1,210,935 comments and 121,027,500 impressions. YouTube has 6,045,720 likes, 2,479,543 shares, 1,235,523 comments and 122,988,419 impressions. Reddit has 5,925,053 likes, 2,364,548 shares, 1,216,551 comments and 116,488,833 impressions. Instagram has 5,901,643 likes, 2,454,870 shares, 1,172,787 comments and 117,664,271 impressions. Twitter has 5,883,120 likes, 2,398,569 shares, 1,216,353 comments and 119,567,039 impressions.

Facebook is on top in likes and has the highest reported average engagement rate, which is 0.29. Instagram also has 0.29; the exact numbers would be needed to make a comparison between the two. YouTube has the total impressions and total comments in this table.

These numbers show both how much people are engaging and how posts each platform has. If the number of posts is different total likes alone is not a way to judge how well a post does on average. A full comparison should include post counts, engagement, average engagement rate and uncertainty along with the totals.

The platform table in the notebook is helpful, for finding places to look deeper. It should not be used to say one platform is always better because the audience size, type of posts, content type how often posts are made and how each platform counts things can be different.

Q2 — ENGAGEMENT BY SENTIMENT

The sentiment analysis puts records into three groups: Negative, Neutral and Positive. It looks at how posts there are the average number of likes the average number of shares the average number of comments and the average engagement rate.

Output: Negative has 4,854 posts, average likes 2,514.73 average shares 1,007.26 average comments 502.29 and average engagement rate 0.28. Neutral has 2,307 posts, likes 2,471.07 average shares 998.41 average comments 502.77 and average engagement rate 0.31. Positive has 4,839 posts, average likes 2,476.00 average shares 1,011.25 average comments 507.16 and engagement rate 0.26.

Neutral posts have the average engagement rate in the recorded output even though they have fewer posts than the positive and negative groups. Negative posts have the average likes while positive posts have the highest average shares and comments. The differences are descriptions and do not show that neutral language leads to more interaction.

One idea to check—not to take for granted—is that content that is controversial or informative might make users react even if the sentiment is positive or negative. To study this correctly future analysis would need to take into account the platform the number of impressions the topic, the campaign phase and the time when the post was made. Sentiment labels could also be wrong if they were made automatically.

Sentiment should be looked at with the type of engagement. A like might mean approval, a notice or just seeing the post while comments can show disagreement. So high engagement does not always mean a reaction, from the audience.

Q2 — BRAND PERFORMANCE

The notebook organizes posts by brand. Shows total likes, shares, comments, average engagement rate and the number of posts for each brand. Ten brands appear in the data: Samsung, Adidas, Microsoft, Google, Pepsi, Apple, Amazon, Nike, Coca-Cola and Toyota.

Samsung has the total likes with 3,073,984. Adidas comes in second with 3,069,471. Microsoft follows with 3,025,517. Samsung also leads in shares with 1,245,855. Microsoft has the average engagement rate at 0.34 when rounded and 0.341 in the merged brand table. Adidas has an average of 0.298 Toyota 0.296 and Nike 0.287 in the merged table.

The brand with the total likes is not the same as the brand with the highest average engagement rate. This difference matters. Total interactions add up across all posts while average engagement rate looks at the performance per post. When comparing brands it's important to consider both the number of posts and how audience exposure each brand has.

Microsoft has 1,239 posts Adidas has 1,229. Samsung has 1,216 in the output. Coca-Cola shows the average engagement rate at 0.228 while Pepsi is at 0.238. These numbers can guide which content to review closely. But the dataset does not include campaign spending, follower counts or verified conversions so it cannot measure return on investment.

Recommendation: Compare brands using both metrics and, per-post measures. Then break the data down by platform, campaign phase, product and topic before deciding which content strategy to copy.

Q3 — MISSING VALUES, DUPLICATES AND QUALITY

The first look at the data shows there are twelve thousand rows and twenty-eight columns. In the mentions column there are three thousand nine hundred forty-one missing values. All other columns have no missing values. The notebook found zero rows. The cleaning process removes rows changes timestamp values using pd.to_datetime with errors set to 'coerce' fills missing values with the median of the column and puts 'Unknown' in missing object columns. After cleaning the data has a shape of twelve thousand by twenty-eight. There are zero missing values and zero duplicate rows. The mentions field might be optional because not every social post mentions another account. Putting 'Unknown' in every mention makes the data complete for work but analysts need to remember the difference between 'no mention' and 'mention data not available' when that difference is important. A accurate way would be to use 'None' for no mention and a different symbol for missing data. The timestamp conversion uses a method that turns dates into NaT. The output says there are no missing values but the notebook should check timestamp.isna().sum() after the conversion. This is because converting to datetime can create missing values that the object column fill step does not fix. Quality checks should also look at duplicates, by post_id check value ranges look for engagement-rate extremes and check category spelling. Checking for duplicate rows and duplicate post identifiers are important. The current notebook shows the cleaning steps. Adding these checks would make the analysis more reliable.

FEATURE ENGINEERING, INDEXING AND RESHAPING

The notebook shows three data-wrangling techniques that go beyond grouping.

First pd.cut divides engagement_rate into Low, Medium and High by using ranges from infinity to 0.05, 0.05 to 0.15 and 0.15 to positive infinity. The labels are Low, Medium and High. Since the upper part of the data has values over 1 and a maximum of 32.21171 these groups are based on the notebooks choices, not rules. The limits of the bins need to be written down and checked against how the rate's defined.

Second the notebook builds a two-level index using platform and sentiment_label. A MultiIndex lets people look at data by than one category like the interactions for a specific platform-sentiment pair. This is helpful for comparisons and short summaries. Groupby tables might be simpler for people who do not know about multi-level indexes.

Third pd.melt changes a part of the data from wide format to long format. Platform is the identifier and likes_count shares_count and comments_count are the variables being measured. Long format is often useful when making plots with the axis and, for tidy-data methods.

The notebook also combines brand engagement data with brand post counts and average sentiment using brand_name. This makes one table that has parts of brand performance. The merge uses a shared key so it is important that brand names are spelled the way. If the names are different it could cause repeated information.

These changes show how preparing data helps with analysis not just fixing it. Whether a feature is useful depends on having a meaning and on whether it keeps the original variables meaning.

Q4 — CAMPAIGN PHASE ANALYSIS

The notebook groups by platform and by campaign phase. The campaign phases are Pre-Launch, Launch and Post-Launch. For each pairing of platform and campaign phase the notebook sums likes, shares, comments and calculates the average engagement rate.

The recorded average engagement rates are: Facebook 0.278 during Launch 0.307 during Post-Launch, 0.275 during Pre-Launch; Instagram 0.235 during Launch 0.345 during Post-Launch 0.279 during Pre-Launch; Reddit 0.250 during Launch 0.221 during Post-Launch, 0.366 during Pre-Launch; Twitter 0.313 during Launch 0.252 during Post-Launch, 0.210 during Pre-Launch; YouTube 0.199 during Launch 0.294 during Post-Launch, 0.340 during Pre-Launch.

The strongest phase differs for each platform. For platform Facebook the highest average engagement rate occurs during Post-Launch. For platform Instagram the highest average engagement rate also occurs during Post-Launch. For platform Reddit the highest average engagement rate occurs during Pre-Launch. For platform Twitter the highest average engagement rate occurs during Launch. For platform YouTube the highest average engagement rate occurs during Pre-Launch. These differences show that campaign phase patterns are not the same across platforms in this data set.

These differences should be interpreted with caution. The groups may differ in the number of posts the type of posts how exposure they received the topic, the audience and the timing. The average engagement rate alone does not prove that a campaign was a success. A stronger evaluation would compare content and add data such, as impressions, spend, reach and business outcomes.

In practice phase-level summaries can help teams ask questions. For example they can ask which formats were used before Launch, what changed during Launch and whether engagement stayed high after Post-Launch. The data can guide follow-up investigation. Causal claims need a more rigorous design.

Q4 — DAILY TIME-SERIES ANALYSIS

The notebook parses timestamp as datetime, removes records with missing timestamps for time analysis, derives a date field and groups engagement_rate by date to calculate a daily average. It also counts posts per date and plots the daily average engagement rate as a line chart.

The recorded peak date is 2024-05-19. On that date, the average engagement rate is 1.6561 across 18 posts. Those posts have 45,343 total likes and 15,714 total shares. Platform counts on the peak date are Twitter 7, Reddit 5, YouTube 3, Instagram 2 and Facebook 1.

An average engagement rate above 1 can be possible only under certain rate definitions or if the metric is not a conventional bounded proportion. Since the notebook's overall maximum is 32.21171, the formula, units and scale must be verified before interpreting the peak as an exceptionally successful day. It may reflect a valid ratio, an unusual record, or a data-generation/calculation issue.

A daily line chart is useful for identifying spikes, troughs and recurring patterns. However, daily averages can be noisy when few posts are present. The post count should be plotted alongside the rate or used to filter days with small samples. A moving average can help distinguish broad trends from day-to-day fluctuation.

The peak-date output provides a concrete candidate for content-level review. The next step would be to inspect the posts, topics and campaign activities on that date, while avoiding assumptions about the reason for the spike.

Q4 — MONTHLY TREND AND DATE FILTERING

The notebook indexes valid timestamps and resamples engagement_rate to month-start frequency (MS), calculating a mean for each month. It also filters records from January 2025 to demonstrate date-based selection.

Recorded monthly averages are: May 2024 0.295055; June 0.225235; July 0.246279; August 0.282617; September 0.270974; October 0.235105; November 0.326860; December 0.258747; January 2025 0.330479; February 2025 0.342387; March 2025 0.228107; and April 2025 0.300749.

Among these monthly values, February 2025 has the highest average at approximately 0.3424, followed by January 2025 at 0.3305 and November 2024 at 0.3269. June 2024 and October 2024 are comparatively lower. These results describe the supplied dataset's monthly means and should not be generalized to all social-media activity.

Monthly aggregation smooths some daily variation, but it can hide differences in post volume and the mix of platforms. A month with a small number of high-rate posts may have a high mean. A more complete report would include monthly post counts, medians, dispersion and platform-adjusted trends.

Date filtering also illustrates the advantage of datetime indexing: analysts can select a calendar month with a string such as 2025-01 rather than manually comparing text dates. The notebook's example returns a sample of January 2025 posts with their platform, engagement rate and likes.

Q4 — CORRELATION ANALYSIS

The notebook computes a Pearson correlation matrix for sentiment_score, toxicity_score, likes_count, shares_count, comments_count, impressions, engagement_rate, user_past_sentiment_avg, user_engagement_growth and buzz_change_rate. A heatmap displays the pairwise values.

The clearest non-diagonal relationship in the displayed matrix is between impressions and engagement_rate, with a correlation of −0.30. Likes_count and engagement_rate have a small positive correlation of 0.08; shares_count and engagement_rate are around 0.04. Most correlations involving sentiment score, toxicity score, past sentiment, engagement growth and buzz change are close to zero in the rounded matrix.

A correlation coefficient ranges from −1 to +1. Values near +1 indicate a strong positive linear relationship; values near −1 indicate a strong negative linear relationship; values near zero indicate little linear association. Near-zero Pearson correlation does not rule out nonlinear relationships or subgroup effects.

The negative impressions-engagement-rate relationship needs special care. If engagement_rate is calculated using impressions as a denominator, a negative association may partly arise mathematically rather than from a behavioral effect. The notebook does not define the rate formula, so the relationship should be checked against the metric definition before making claims.

Correlation does not imply causation. The matrix cannot establish that increasing impressions reduces engagement. Confounding factors, denominator effects, outliers and platform-specific definitions can influence the observed values. Future work should validate the metric and examine scatter plots, robust correlations and segmented analyses.

Q5 — VISUALIZATION STRATEGY

The notebook uses several visualization types because each answers a different question. Bar plots compare totals across categories; line plots show change over time; heatmaps summarize a two-way table or correlation matrix; pie charts show shares of a whole; histograms reveal distributions; scatter plots show relationships between two numeric variables; and box plots compare distributions across groups.

Visualizations in the notebook include total likes by platform, total comments by platform, average engagement by platform and sentiment, post counts by sentiment, share of posts by platform, total likes by brand, engagement-rate distribution, impressions versus likes, sentiment scores by platform and toxicity by sentiment label.

Good chart design requires meaningful titles, clear axis labels, readable category names and appropriate scales. For platform comparisons, bars should use a consistent scale. For rate charts, the unit and calculation should be stated. For pie charts, the categories should represent a complete whole and should not be too numerous.

Visuals are evidence aids rather than proof. A scatter plot can show a pattern, but the relationship should also be summarized numerically and examined for outliers. A heatmap can make correlations easier to scan, but values should be read directly rather than inferred only from color. A line chart should not imply continuous measurements between dates if observations are sparse.

The notebook's plots provide an accessible summary of a complex dataset and make it easier to spot patterns worth investigating further.

FIGURE 1 — CORRELATION HEATMAP

The correlation heatmap displays the relationships among ten numerical fields. The diagonal is 1.00 because each variable is perfectly correlated with itself. Off-diagonal values show the linear association between two different fields.

The most noticeable reported off-diagonal value is approximately −0.30 between impressions and engagement_rate. A smaller positive value of 0.08 appears between likes_count and engagement_rate. Most other values are close to zero when rounded to two decimal places.

Interpretation should prioritize magnitude and context. A correlation of −0.30 is not a perfect inverse relationship; it is a modest negative linear association. A correlation near zero indicates little linear association in the full sample, but associations may vary across platforms, sentiment categories or campaign phases.

The heatmap is useful because it condenses many pairwise comparisons into one graphic. However, colors can exaggerate small differences if the scale is not fixed. The notebook fixes the color scale from −1 to +1, which makes the values comparable across cells.

Figure: the heatmap below is extracted from the submitted notebook output.

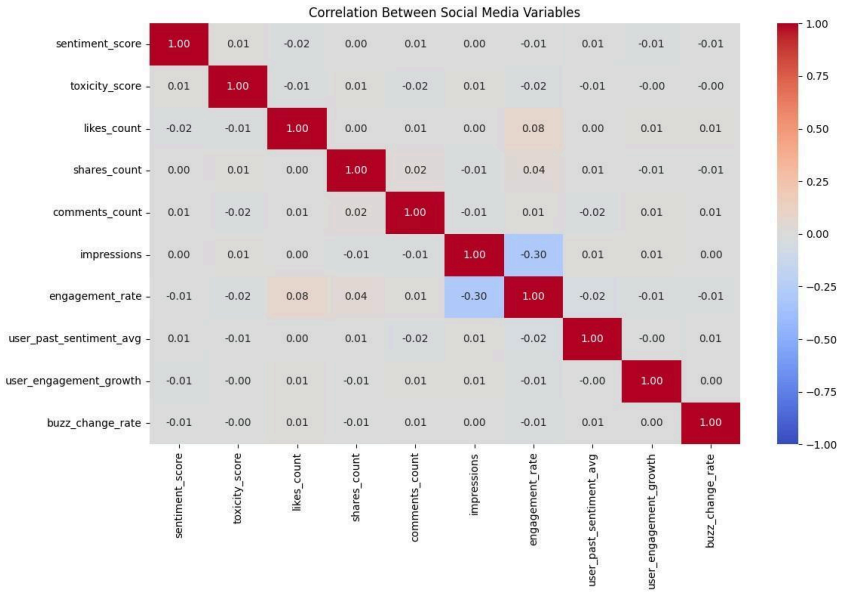


Figure reproduced from the submitted Colab notebook output.

FIGURE 2 — DAILY ENGAGEMENT TREND

The daily line plot shows the average engagement_rate for each date. A point represents the mean of the posts recorded on that date. Connecting points helps reveal changes through the observation period.

The notebook identifies 19 May 2024 as the date with the highest daily average engagement rate (1.6561). Eighteen posts occur on that date, with 45,343 likes and 15,714 shares combined. Twitter contributes seven posts and Reddit five, followed by YouTube, Instagram and Facebook.

The peak is a descriptive observation, not an explanation. To understand it, one would inspect the post text, topics, hashtags, campaign phases and impressions for that date. It is also important to check whether the unusually high engagement-rate values arise from the metric formula or outliers.

Daily series can be sensitive to uneven post counts. The interpretation is stronger when daily post count is shown in a second panel or when the chart is supplemented with a rolling average. The chart is retained here as a faithful illustration of the notebook's workflow.

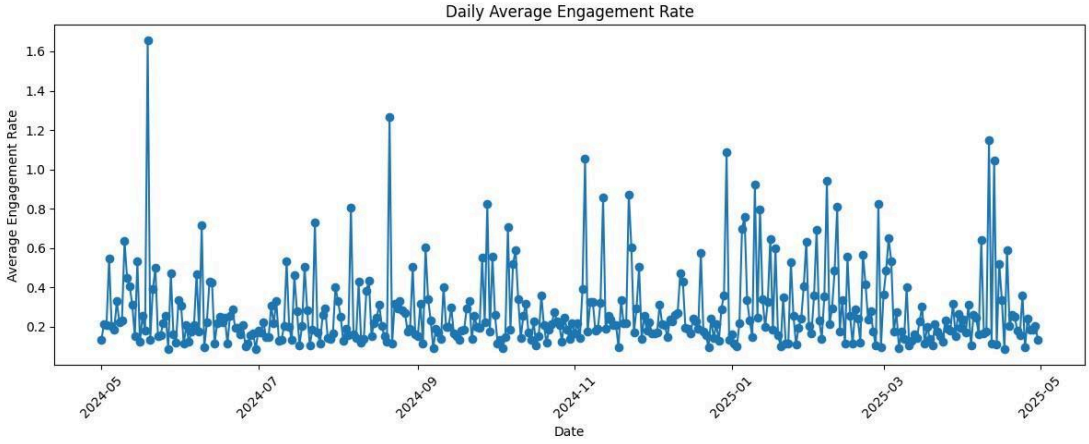


Figure reproduced from the submitted Colab notebook output.

FIGURE 3 — MONTHLY ENGAGEMENT TREND

The monthly line plot resamples timestamp-indexed engagement_rate to monthly means. It provides a less granular view than the daily chart and makes broader month-to-month variation easier to inspect.

The notebook's monthly averages rise from 0.2252 in June 2024 to 0.3269 in November 2024, fall to 0.2587 in December, and rise again to 0.3305 in January 2025 and 0.3424 in February 2025. March 2025 drops to 0.2281 before April reaches 0.3007.

These changes indicate variation across months in the supplied records, but they do not establish a seasonal pattern. The observation window is only twelve months, and differences in the number of posts, platform mix, campaign activity or content topics could explain part of the movement.

Future analysis could overlay monthly post counts and compare platform-specific trends. A longer period would help distinguish recurring seasonality from isolated variation. The date index also makes it straightforward to filter and compare specific months.

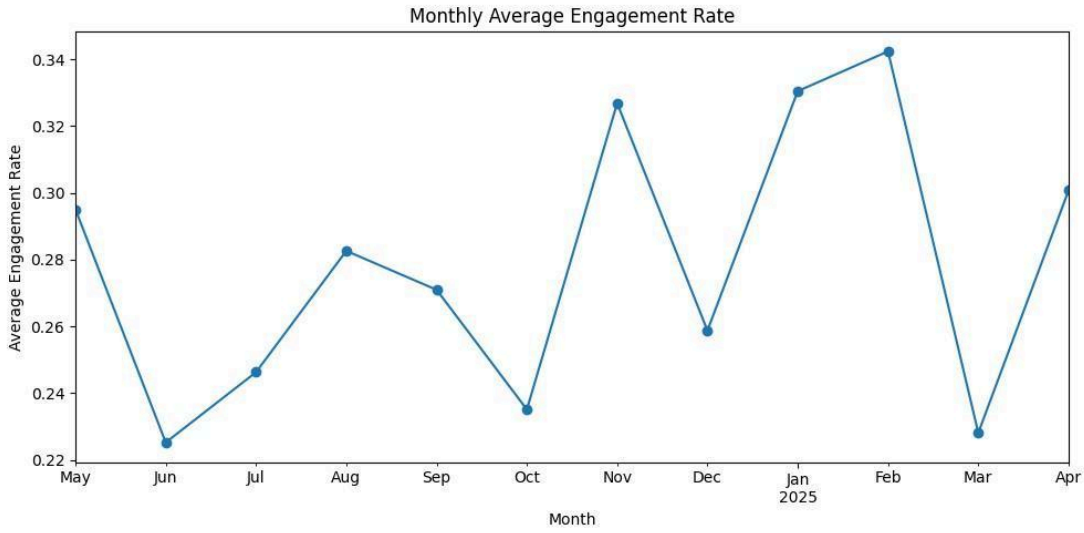


Figure reproduced from the submitted Colab notebook output.

FIGURE 4 — TOTAL LIKES BY PLATFORM

The bar chart of total likes compares the accumulated likes across the five platforms. Facebook leads with 6,133,107 total likes. YouTube follows with 6,045,720; Reddit has 5,925,053; Instagram has 5,901,643; and Twitter has 5,883,120.

The difference between Facebook and Twitter is approximately 250,000 likes across the entire dataset. These are cumulative totals, so they reflect both the number of posts and the likes received by each post. The chart should not be interpreted as an estimate of average post quality.

Pairing this chart with post counts and average engagement rates provides a more balanced comparison. For example, YouTube has the highest total impressions and comments in the platform table, while Facebook has the highest total likes. Different measures can therefore identify different forms of platform performance.

The notebook uses a Seaborn bar plot with the sum of likes as the estimator. This is appropriate for a total comparison, provided the audience understands that the statistic is a sum rather than a mean.

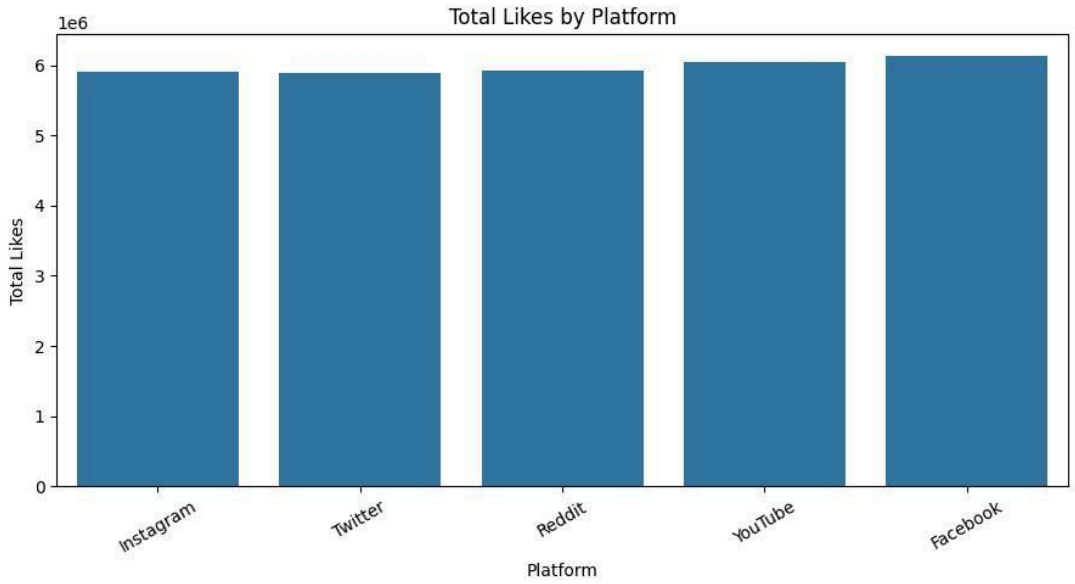


Figure reproduced from the submitted Colab notebook output.

FIGURE 5 — TOTAL COMMENTS BY PLATFORM

The total-comments chart compares the number of comments recorded for each platform. The platform table shows YouTube at 1,235,523 comments, Reddit at 1,216,551, Twitter at 1,216,353, Facebook at 1,210,935 and Instagram at 1,172,787.

Comments can represent discussion, questions, feedback, disagreement or spam. A higher count is therefore not automatically positive. For campaign analysis, comments may need to be categorized by topic, sentiment, intent and moderation status.

The total is also affected by posting volume. To compare how likely a post is to prompt discussion, the analysis could calculate average and median comments per post, comments per impression, and the share of posts with no comments.

The notebook's bar chart makes cross-platform totals easy to compare. The discussion illustrates a general principle of EDA: metrics should be interpreted according to what they measure, and a single metric rarely captures the full meaning of engagement.

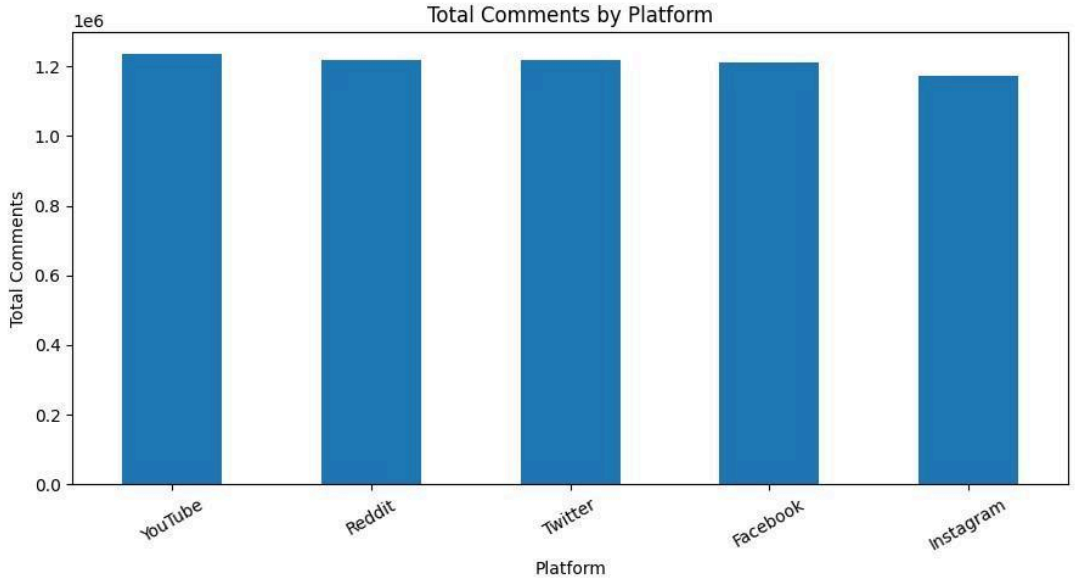


Figure reproduced from the submitted Colab notebook output.

FIGURE 6 — PLATFORM AND SENTIMENT HEATMAP

The platform-versus-sentiment heatmap uses a pivot table with platform as rows, sentiment_label as columns and mean engagement_rate as the cell value. This chart allows the analyst to see whether the sentiment pattern differs by platform.

A heatmap can reveal interactions that are hidden in an overall sentiment average. For example, a sentiment category may show relatively high average engagement on one platform but not on another. The notebook creates this table directly from the post-level dataset rather than from manually entered values.

Cell values should be read alongside the number of posts in each platform-sentiment group. A high mean based on few records may be less stable than a similar mean based on many records. The current plot displays means, while group counts would provide useful supporting context.

Sentiment analysis is not equivalent to audience approval. Negative sentiment may still produce high interaction because users debate or criticize the content. Therefore, engagement and sentiment should be analyzed together, not treated as interchangeable measures.

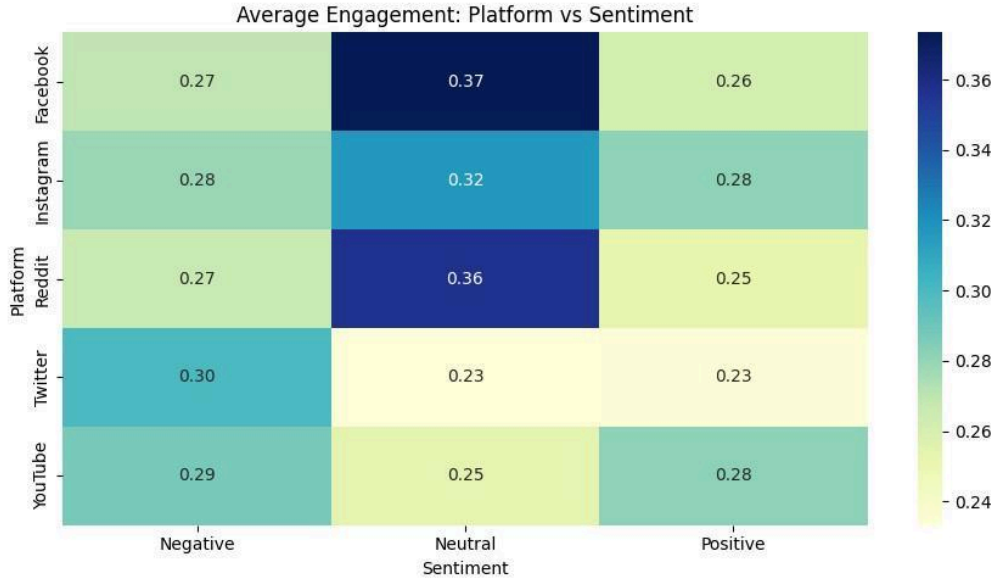


Figure reproduced from the submitted Colab notebook output.

FIGURE 7 — SENTIMENT DISTRIBUTION

The sentiment distribution chart counts posts by label. The recorded sentiment table shows 4,854 Negative posts, 4,839 Positive posts and 2,307 Neutral posts, for 12,000 records in total.

Negative and positive posts are nearly equal in number, while neutral posts are less common. This balance helps contextualize the average sentiment score, which is close to zero. The overall mean can be near zero when positive and negative values offset each other.

Counts should be distinguished from engagement performance. A category can contain many posts without having the highest average engagement rate. The earlier sentiment summary reports the Neutral category's average engagement rate as 0.31, compared with 0.28 for Negative and 0.26 for Positive.

The bar chart is preferable to a pie chart for comparing three counts because differences in bar height are easier to judge accurately. The notebook labels the categories and uses a clear y-axis to communicate the distribution.

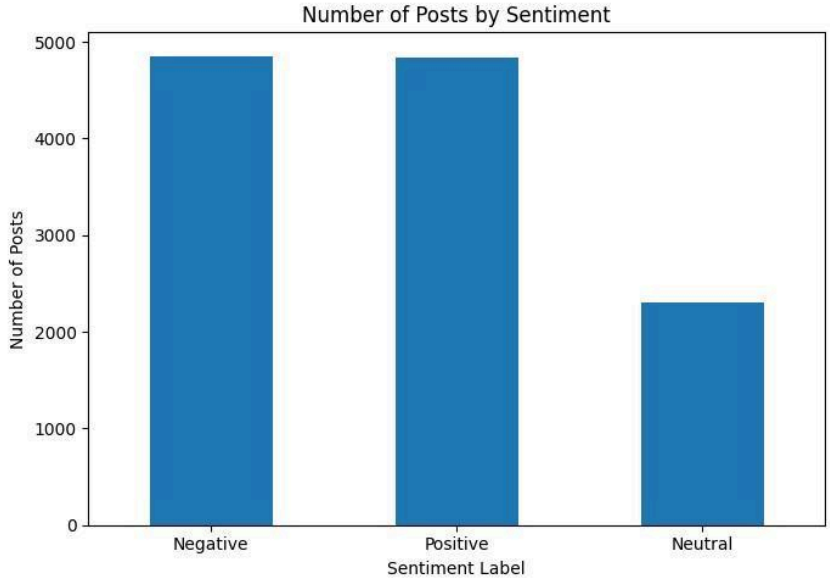


Figure reproduced from the submitted Colab notebook output.

FIGURE 8 — SHARE OF POSTS BY PLATFORM

The pie chart shows each platform's share of all posts in the dataset. The five platforms are Facebook, YouTube, Reddit, Instagram and Twitter. Because the notebook uses value_counts(), the slices represent post counts rather than likes, impressions or engagement rate.

A pie chart is useful when the purpose is to show composition of a whole. The categories should add up to 100 percent, and the chart should not be read as a measure of platform popularity outside this sample.

Platform post shares are important context for interpreting total interactions. If one platform contains more records, its cumulative likes may be higher even if its typical post receives fewer likes. The dataset's platform totals are relatively close, but a normalized comparison remains valuable.

Alternative displays include a sorted bar chart with percentages and a table of post counts. These are often easier to compare when slice sizes are similar. The notebook's pie chart demonstrates a standard visualization option for categorical proportions.

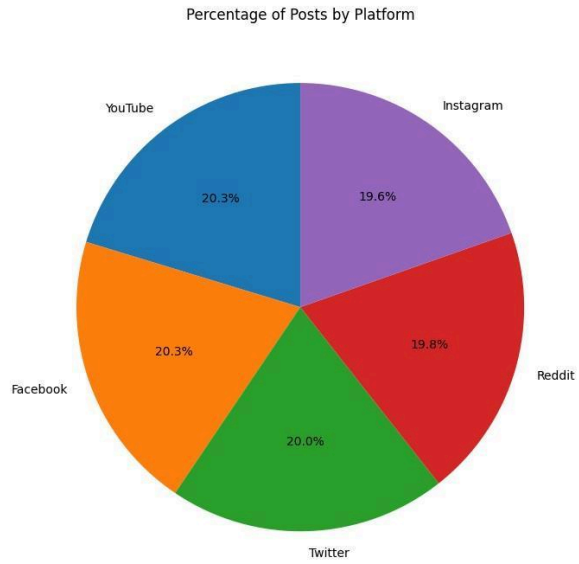


Figure reproduced from the submitted Colab notebook output.

FIGURE 9 — BRAND ENGAGEMENT

The brand bar chart compares total likes by brand. Samsung leads with 3,073,984 likes, Adidas follows with 3,069,471 and Microsoft has 3,025,517. The remaining brands have totals in a relatively narrow range, with Toyota at 2,889,510 in the recorded table.

The brand analysis also reports average engagement rate and post count. Microsoft has the highest average engagement rate at 0.341, while Samsung leads in total likes. This difference reinforces why both totals and rates are needed.

Brand performance can be influenced by how many posts were published, which platforms were used, content topics, campaign phases, audience size and the distribution of impressions. These factors are not all controlled for in the notebook's simple grouping.

The chart should be treated as a screening tool. A deeper comparison would calculate per-post interaction distributions, rates normalized by exposure, confidence intervals and results by platform. Campaign spend and conversion outcomes would be required to assess business return.

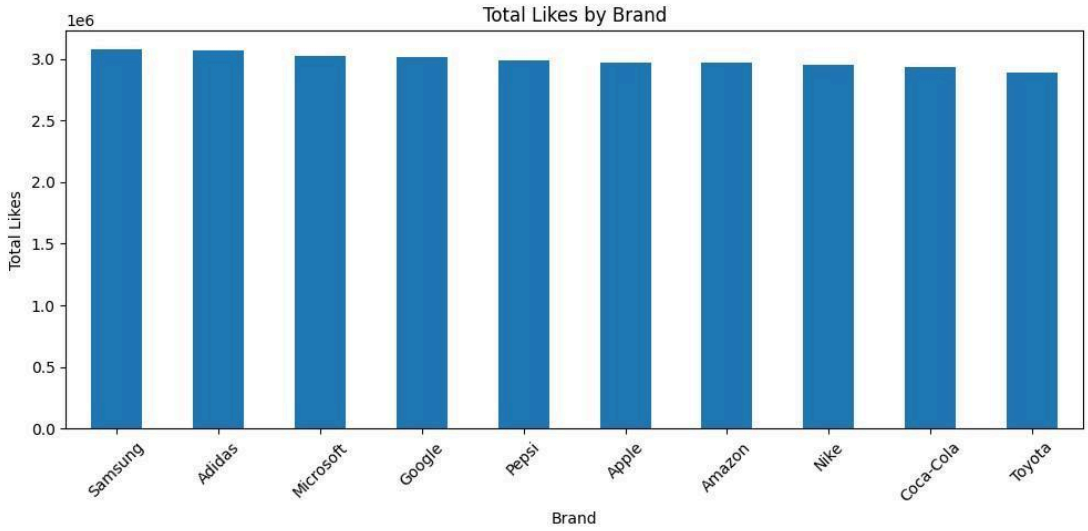


Figure reproduced from the submitted Colab notebook output.

FIGURE 10 — ENGAGEMENT-RATE DISTRIBUTION

The histogram plots engagement_rate with twenty bins and a kernel density estimate. The notebook's descriptive statistics show a mean of 0.278137, median of 0.080605, 75th percentile of 0.163123 and maximum of 32.211710.

The mean is substantially larger than the median and the maximum is far above the upper quartile. This indicates a right-skewed distribution with a long upper tail. A small number of extreme observations may pull the mean upward.

Because the rate definition is not included in the notebook, it is not possible to determine from the chart alone whether values above 1 are valid ratios, percentage points, or calculation anomalies. Before using the measure, document its formula, numerator, denominator and scale. Check extreme rows and verify whether impressions can be zero or unusually small.

Useful follow-up summaries include median and interquartile range by platform, percentile-based plots, and a histogram on a carefully chosen scale. Outliers should be investigated rather than automatically removed.

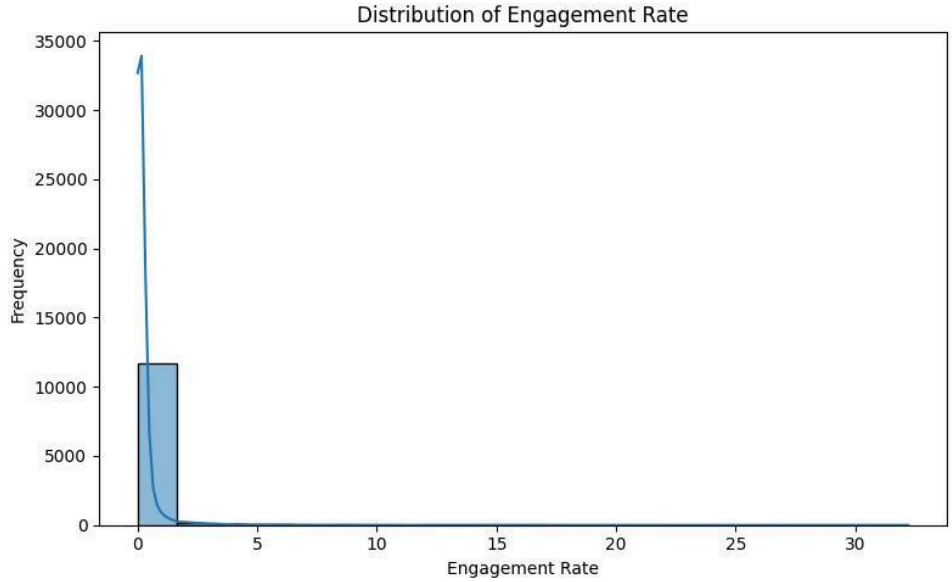


Figure reproduced from the submitted Colab notebook output.

FIGURE 11 — IMPRESSIONS VERSUS LIKES

The scatter plot places impressions on the horizontal axis and likes_count on the vertical axis, with points distinguished by platform. It helps show whether posts with more exposure tend to receive more likes and whether the pattern differs across platforms.

The correlation matrix reports a correlation of approximately 0.00 between impressions and likes_count when rounded to two decimal places. This indicates little linear association in the displayed full-sample Pearson correlation, despite the intuitive possibility that more impressions could create more opportunities for likes.

Potential reasons include the way the synthetic or supplied data were generated, variation in content appeal, platform differences, and the fact that the plot compares raw likes rather than likes per impression. The plot may also contain overlapping points, which can obscure density.

Future work could calculate likes per impression, use transparency or hexbin density plots, and analyze within each platform. If engagement_rate uses impressions in its denominator, that formula should be documented before interpreting its relationship with impressions.

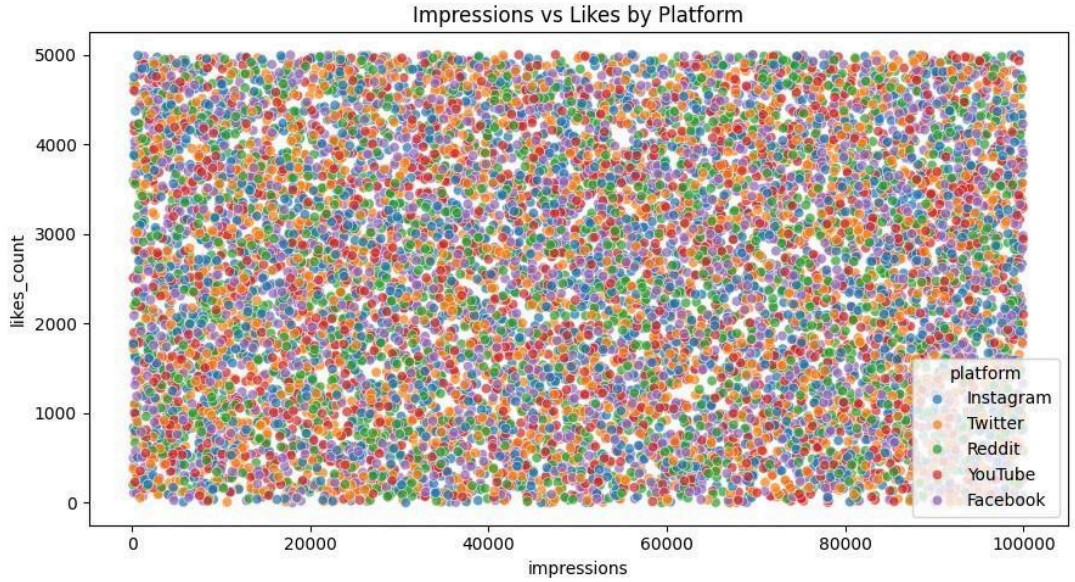


Figure reproduced from the submitted Colab notebook output.

FIGURE 12 — SENTIMENT SCORES BY PLATFORM

The box plot compares the distribution of sentiment_score across platforms. A box plot displays the median, interquartile range, whiskers and potential outliers, allowing differences in central tendency and spread to be inspected.

The notebook's full-sample sentiment_score mean is 0.000553, and its range is approximately −1 to +1. This near-zero overall mean does not mean every post is neutral; it can arise from a mixture of positive and negative observations.

Platform-specific box plots help check whether one platform has a different median sentiment or more variable scores. Any visible differences should be described in terms of the distribution rather than assumed to reflect the platform itself. Content topics, language, user populations and campaign composition may differ.

Because sentiment scores may come from an automated model, validation on a manually reviewed sample would improve confidence. Sarcasm, slang, mixed-language posts and domain-specific terminology can affect sentiment classification.

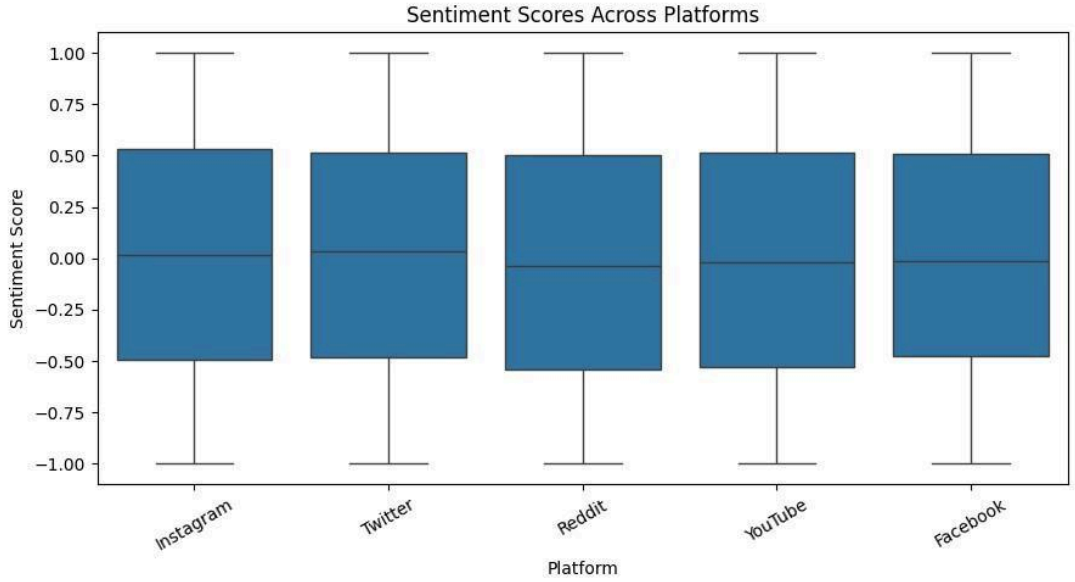


Figure reproduced from the submitted Colab notebook output.

FIGURE 13 — TOXICITY BY SENTIMENT LABEL

The final notebook plot compares toxicity_score across Negative, Neutral and Positive sentiment labels using a box plot. It examines whether the distributions of a toxicity indicator differ by the sentiment category.

Toxicity and sentiment are related but distinct concepts. A negative post may be respectful criticism, while a positive-sounding post can contain harassment or sarcasm. A toxicity score should not be treated as a definitive judgement without understanding the model, threshold and context.

The recorded overall toxicity_score mean is 0.503868, with values from 0 to approximately 1. The notebook does not provide a detailed definition of this score, so the report avoids assigning an external meaning to specific thresholds.

For moderation or safety decisions, scores should be combined with human review, policy definitions and appeal mechanisms. The plot is useful for exploratory comparison, but should not be used alone to label individuals or remove content.

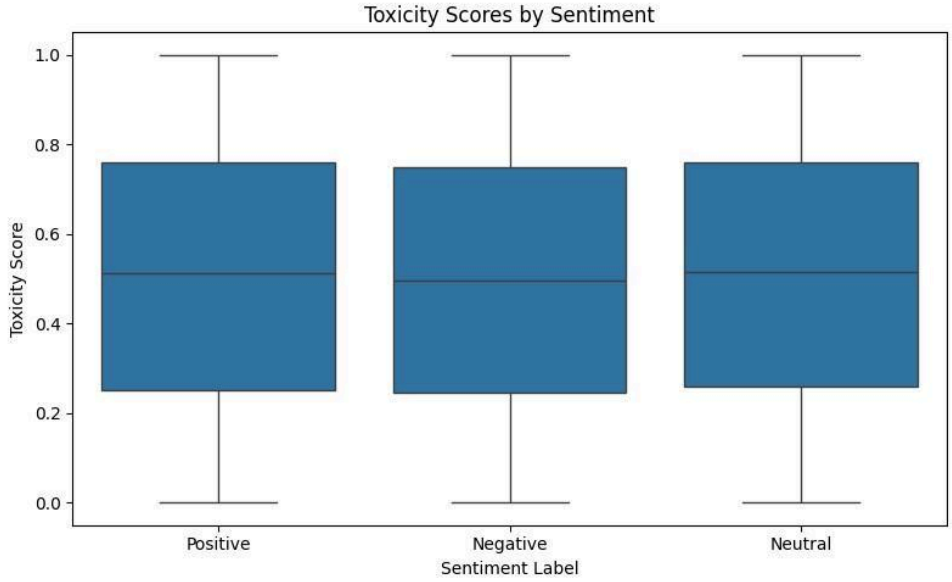


Figure reproduced from the submitted Colab notebook output.

RESULTS, DISCUSSION AND KEY FINDINGS

The notebook supports the following descriptive findings:

1. Dataset size: 12,000 rows and 28 columns.

2. Data quality: 3,941 missing mentions initially; zero exact duplicate rows; after cleaning, zero missing values and zero duplicatesare reported.

3. Platform likes: Facebook has the largest total likes (6,133,107) and the highest average likes per post (2,522.87).

4. Platform exposure: YouTube has the largest total impressions (122,988,419) and total comments (1,235,523) in the platformsummary.

5. Sentiment: Neutral posts have the highest average engagement rate (0.31), while negative and positive posts are more numerous.

6. Brand: Samsung has the highest total likes (3,073,984); Microsoft has the highest average engagement rate (0.341).

7. Time: 19 May 2024 is the peak daily average engagement date (1.6561) across 18 posts; February 2025 has the highest monthlyaverage among the displayed months (0.342387).

8. Correlation: impressions and engagement_rate have a displayed correlation of −0.30; most other correlations are near zero.

These findings answer the notebook's questions while also revealing important validation needs. In particular, the engagement-rate distribution has a high maximum, and the metric formula is not documented. The findings are valid descriptions of the recorded outputs, not causal explanations or general truths about the platforms.

LIMITATIONS, RECOMMENDATIONS AND FUTURE SCOPE

Limitations: the report is based on one supplied dataset and notebook output; the CSV was not separately attached for independent row-level validation. The time window spans approximately May 2024 to April 2025. The engagement-rate formula is undocumented, and the distribution contains extreme values. The dataset does not provide verified audience sizes, advertising costs, conversions or business outcomes. Sentiment and toxicity scores may be model-generated and can contain classification errors.

Recommendations:

- Document the engagement-rate formula and verify units before interpreting values above 1.
- Inspect extreme engagement-rate records, timestamps and post identifiers; do not remove outliers without evidence.
- Report post counts alongside totals and average rates for every platform, brand and campaign phase.
- Use medians, quartiles and robust summaries in addition to means.
- Segment platform results by content topic, campaign phase, date and sentiment.
- Review high-engagement posts manually to understand context rather than assuming a cause.
- Validate automated sentiment and toxicity labels on a representative sample.

Future scope: add follower counts, reach, paid spend, click-throughs, conversions and content-format fields; build an interactive dashboard; conduct statistical tests with suitable assumptions; compare longer time periods; use robust or rank-based correlation; and design controlled experiments for causal campaign evaluation.

Responsible use: aggregated findings should guide content improvement, not profile individual users or make high-impact decisions from sentiment/toxicity scores alone.

CONCLUSION

This course-end project demonstrates an end-to-end exploratory data analysis workflow for social media engagement. Using NumPy, Pandas, Matplotlib and Seaborn, the notebook inspects 12,000 post records, checks data quality, summarizes engagement across platforms and brands, compares sentiment groups and campaign phases, analyzes daily and monthly changes, calculates correlations and produces a collection of visualizations.

The recorded outputs identify Facebook as the platform with the highest total likes, Samsung as the brand with the highest total likes, and Microsoft as the brand with the highest average engagement rate. Neutral sentiment posts have the highest average engagement rate among sentiment labels. The time-series analysis identifies 19 May 2024 as the peak daily average engagement date, while February 2025 has the highest monthly mean among the months shown.

The most important analytical caution concerns the engagement_rate field. Its mean is 0.278137, median is 0.080605 and maximum is 32.21171. These values suggest a long right tail and require the rate definition to be confirmed before making business comparisons. The negative correlation between impressions and engagement rate should also be interpreted in light of the unknown formula.

Overall, the project shows how EDA turns raw records into structured evidence. Its strongest contribution is not merely ranking platforms or brands, but demonstrating a transparent workflow and identifying what should be validated next. More contextual data, robust statistics and outcome measures would support stronger conclusions in future work.

REFERENCES

1. NumPy Developers. NumPy documentation. https://numpy.org/doc/

2. The pandas development team. pandas documentation. https://pandas.pydata.org/docs/

3. Matplotlib Development Team. Matplotlib documentation. https://matplotlib.org/stable/

4. Waskom, M. et al. seaborn documentation. https://seaborn.pydata.org/

5. Google Colaboratory documentation. https://colab.research.google.com/

6. Submitted course project notebook: HansikaEDAVProject(2).ipynb (provided by the student).

7. Input data file referenced in the notebook: Social_Media_Engagement_Dataset.csv. The raw CSV was not separately attached forthis report.

APPENDIX A — NOTEBOOK IMPLEMENTATION SUMMARY

Core imports: import numpy as np; import pandas as pd; import matplotlib.pyplot as plt; import seaborn as sns.

Loading: df = pd.read_csv('Social_Media_Engagement_Dataset.csv'). The notebook uses the Google Colab upload utility to provide the CSV file.

Data inspection: df.shape, df.columns, df.info(), df.describe(), df.isnull().sum() and df.duplicated().sum().

Cleaning: df.drop_duplicates(); pd.to_datetime(df['timestamp'], errors='coerce'); median imputation for numeric columns; and 'Unknown' for missing object columns.

Platform statistics: filter rows by platform, convert likes_count to NumPy arrays and compute np.mean, np.min and np.max.

Aggregation: df.groupby(...).agg(...) calculates totals and means by platform, sentiment, brand and campaign phase.

Time series: derive date from timestamp; group daily engagement; resample monthly means with resample('MS').mean(); identify daily peak with idxmax().

Correlation: select numeric columns and call .corr(); visualize with sns.heatmap().

Other techniques: pd.cut for engagement categories, MultiIndex through set_index, pd.melt for reshaping, and pd.merge for combining brand summaries.

Visualization: bar plots, pie chart, histogram with KDE, scatter plot, line plots, and box plots. The original notebook remains the source of executable code.

APPENDIX B — GLOSSARY AND INTERPRETATION NOTES

Engagement: observable interactions with a post, such as likes, shares and comments. The dataset also supplies an engagement_rate field whose formula should be verified.

Impressions: recorded exposures or displays of content. Impressions are not necessarily unique people.

Sentiment score: a numeric measure intended to represent polarity or sentiment strength. The precise model and scale should be documented.

Sentiment label: categorical label such as Positive, Neutral or Negative.

Toxicity score: a numeric indicator related to potentially harmful or abusive content; its definition depends on the model or source.

Mean: arithmetic average. It can be influenced by extreme values.

Median: middle value after sorting; it is more robust to outliers than the mean.

Correlation: a numerical measure of association. Pearson correlation measures linear association and does not establish causation.

Groupby aggregation: summarizing records by one or more categories, such as platform or brand.

Resampling: grouping datetime-indexed observations into time intervals such as months.

Outlier: an observation far from much of the distribution. An outlier may be valid and should be investigated before removal.

Reproducibility: the ability to repeat an analysis from the same data, code and settings and obtain the same results.

APPENDIX C — RECOMMENDED VALIDATION CHECKLIST

Before using the findings for business decisions, verify the following items.

1. Confirm the source, licence, collection method and date range of the CSV file.
2. Confirm that post_id is unique and inspect repeated user_id values only for aggregate patterns that are appropriate to analyze.
3. Check timestamp parsing failures and confirm that day_of_week agrees with timestamp-derived weekdays.
4. Define engagement_rate explicitly: specify numerator, denominator, scale (fraction or percentage), and treatment of missing/zerodenominators.
5. Inspect the highest and lowest engagement-rate rows and verify their underlying likes, shares, comments and impressions.
6. Confirm whether likes, shares and comments are cumulative values, snapshots, or values captured at a single time.
7. Compare means with medians and quantiles for platform and brand groups.
8. Include group sample sizes with all averages and rates.
9. Review sentiment and toxicity model documentation, thresholds and known limitations.
10. Validate a sample of automated labels with human reviewers.
11. Use platform-specific comparisons when platform metrics are defined differently.
12. Document all cleaning decisions, especially imputation and treatment of missing mentions.
13. Keep code, data version and package versions together so the analysis can be reproduced.
14. Avoid causal language unless an appropriate experimental or quasi-experimental design supports it.

APPENDIX D — SAMPLE REPORTING TABLES AND FORMULAS

Platform comparison should ideally include: platform; number of posts; total likes; median likes per post; total shares; median shares; total comments; median comments; total impressions; mean engagement rate; median engagement rate; and missing-data counts.

Brand comparison should include: brand; post count; total likes; likes per post; average and median engagement rate; average sentiment; platform distribution; campaign-phase distribution; and verified outcome measures if available.

Common descriptive formulas:

Mean = sum of observed values divided by number of observed values.

Engagement rate = use the dataset's documented formula. Do not assume a formula solely from the field name.

Pearson correlation r = covariance of X and Y divided by the product of their standard deviations.

Daily mean engagement = mean of engagement_rate among records assigned to a particular calendar date.

Monthly mean engagement = mean of engagement_rate among records assigned to a calendar month.

Percentage share of posts = posts in a category divided by total posts multiplied by 100.

These formulas explain the type of calculation used. The engagement-rate formula remains intentionally unspecified because the notebook does not define it.